Not Just Scanner Bias: A Layered Audit of a Deep Brain-Age Model Beyond Accuracy

Adharshan N*

School of Computer Science and Engineering

Vellore Institute of Technology

Chennai, India

adharshan2706@gmail.com

Jaaiwanth K*

School of Computer Science and Engineering

Vellore Institute of Technology

Chennai, India

jaaiwanth90@gmail.com

Shridevi S†

Centre for Neuroinformatics

Vellore Institute of Technology

Chennai, India

[faculty email]

**Equal contribution. †Corresponding author.*

***Abstract*—Brain-age models predict age from structural MRI, and site disparities in the brain-age gap are often attributed to scanner differences without a test. We audit a UK Biobank-pretrained SFCN fine-tuned on 485 T1-weighted IXI scans from three London sites, with an equivalence margin and verdict rules fixed after the test-set analysis and before cross-validation. The model is accurate (test MAE 3.71 years) and its errors are equivalent between sexes. Scans from one site (IOP) receive a brain-age gap 5.24 years lower than the reference site in 5-fold cross-validation (95% CI −7.56 to −2.92). The offset survives eight ways of handling age and is already present in the pretrained model before any training on IXI (−3.3 years), including away from that model's 42-year output floor. Fine-tuning hides the offset for the subjects it trains on, but not for unseen subjects. The model is sensitive to acquisition appearance: site is decodable from its features, and giving Hammersmith brains IOP's intensity profile lowers their predicted age by 3.9 years. Yet this sensitivity does not explain the disparity. Intensity, smoothness, head size, recorded cohort characteristics and age-conditional removal of site from the features all leave the offset in place; naive concept erasure removes it only by distorting IOP's predictions. Attribution maps pass randomisation and deletion tests. The model's own 90% ranges cover only 77% of out-of-sample IOP subjects, and per-site conformal intervals restore coverage. We summarise the audit as a fairness passport with graded verdicts.**

***Index Terms*—Brain age, MRI, fairness, site effects, shortcut learning, explainable AI, harmonisation, conformal prediction.**

I. INTRODUCTION

Brain-age models learn to predict a person's chronological age from a structural MRI scan [1], [2]. The brain-age gap (BAG) between predicted and chronological age is interpreted as accelerated or delayed brain ageing and has been associated with mortality and neurological disease [3]. Deep convolutional networks predict age with errors of a few years [4]–[7].

Accuracy alone does not make a model trustworthy. A model can be accurate on average and biased for a subgroup [8]–[11]; it can exploit non-biological shortcuts such as hospital-specific acquisition properties [12]–[14]; and the explanations used to inspect it can themselves be unfaithful [15], [16]. In multi-site MRI, scanner differences are a well-known source of site effects [17]–[19], and differences in brain-age gap between datasets can largely reflect their age distributions [20].

These concerns meet in a common situation: a brain-age model shows a disparity between sites, and the disparity is attributed to the scanner. That attribution is rarely tested. We audit a widely used network on the public IXI dataset and treat "scanner bias" as a hypothesis to be tested rather than assumed. Our narrative follows the evidence: (1) the model is accurate; (2) it shows a replicated site disparity that is not an artefact of age handling and exists before fine-tuning; (3) the model is sensitive to acquisition appearance; (4) but that sensitivity does not explain the disparity; (5) a disparity should therefore not be labelled scanner bias without such tests.

Our contributions are:

- A layered audit protocol (Fig. 1) with graded verdicts based on equivalence tests, minimum detectable effects and replication, released as a reusable passport schema in the spirit of model cards [21].
- A site disparity shown to be robust to eight age-handling choices, to cross-validation over all non-test subjects and to independent training runs, and present in the pretrained model before any exposure to IXI.
- Direct evidence of acquisition sensitivity (site decodability against a permutation null; an age- and sex-standardised intensity-swap stress test), and a systematic rule-out showing that neither acquisition properties we can measure nor recorded cohort characteristics explain the disparity.
- A methodological warning: concept erasure that ignores the age–site confound appears to remove the disparity by distorting the predictions of the younger site.
- Validated explanations and uncertainty: attribution maps tested under two deletion baselines and against Integrated

Gradients, and conformal intervals with per-site (Mondrian) coverage.

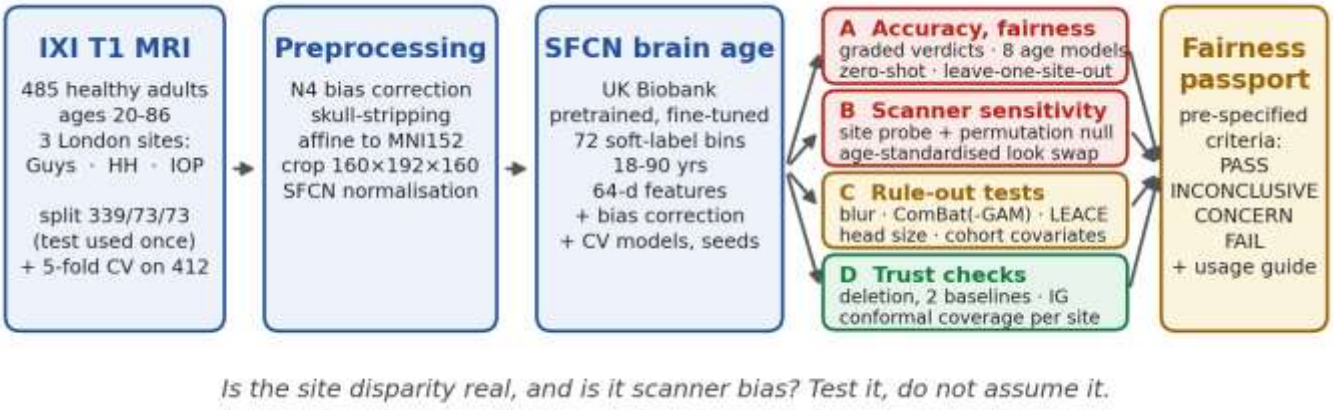


Fig. 1. Overview of the audit. The fine-tuned SFCN is frozen and audited on held-out subjects; cross-validated models over all non-test subjects provide the primary estimates for group effects, and four audit layers are summarised as a protocol-based fairness passport.

II. RELATED WORK

Brain-age prediction. Deep 3D CNNs trained on raw images [4], DeepBrainNet [6], the lightweight SFCN [5] and its multi-cohort descendants [7] predict age within a few years. Brain-age estimates regress towards the mean age, so a bias correction is standard; whether to correct the prediction or include age as a covariate remains debated [22]–[24].

Fairness and shortcuts in medical imaging. Imbalanced data produce biased classifiers [8] and under-served groups are underdiagnosed [9]; see [10]. For brain age, race and sex bias was analysed in UK Biobank [11], and a multi-cohort evaluation found dataset differences largely explained by age distributions, with a small ethnicity effect in some models [20]. Models encode protected attributes they were not trained on [25], measurable with linear probes [26]; decodability does not imply use, which motivates erasure-based tests [27].

Site effects and harmonisation. Scanner signatures are learnable [17]. ComBat [18], [28] and its nonlinear-age extension ComBat-GAM [29] remove site effects from derived measures; adversarial unlearning removes them during training [19], [30]; histogram matching standardises intensity [31].

Faithful explanations and uncertainty. Grad-CAM [32], HiResCAM [33] and Integrated Gradients [34] are common attribution methods; they should pass randomisation [15] and perturbation tests [16], [35]. Network confidence is often miscalibrated [36]; deep ensembles [37] and conformal prediction [38], [39], including group-conditional (Mondrian) variants, provide checked or guaranteed coverage.

III. DATA

IXI [40] provides T1-weighted MRI and demographics for healthy adults scanned at Guy's Hospital (Guys; Philips 1.5 T), Hammersmith Hospital (HH; Philips 3 T) and the Institute of Psychiatry (IOP; GE 1.5 T). Labels were rebuilt from the official spreadsheet after an identifier error in an early table was found; two independent rebuilds agreed, and no result based on the erroneous labels was reported. Of 502 subjects with a scan and a demographic index entry, 17 were excluded (10 without a record, 2 without age, 2 with contradictory duplicates, and 3 with clean labels that were not part of the shared team split; the list is released). The remaining 485 were split 339/73/73 into train, validation and test, stratified by site (Table I). IOP recruited younger participants (Fig. 2), so every group comparison adjusts for age, and Section V-C tests whether that adjustment is adequate.

TABLE I
Subjects per acquisition site

Site	Scanner	n	Mean age	Test n
Guys	Philips 1.5 T	275	50.6	42
HH	Philips 3 T	149	47.7	22
IOP	GE 1.5 T	61	42.5	9

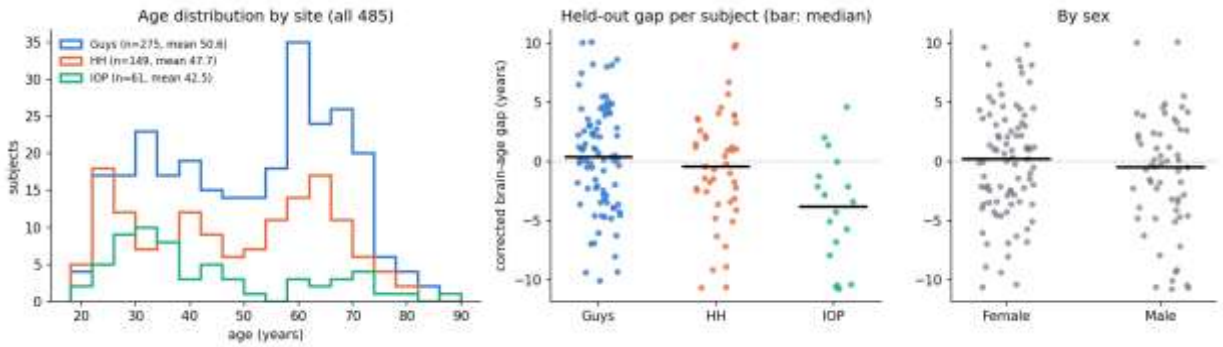


Fig. 2. Left: age distribution by site (all 485 subjects). Middle and right: corrected brain-age gap of every held-out subject by site and by sex (bar: median).

IV. METHODS

A. Preprocessing and model

Each scan underwent N4 bias-field correction [41], skull-stripping with ANTsPyNet [42], affine registration to the MNI152NLin6Asym 1 mm template [43], a centre crop to 160×192×160 voxels and SFCN's intensity normalisation (division by the mean over the full 182×218×182 box). SFCN [5] has five convolution–batch-norm–max-pool–ReLU blocks and a 1×1×1 block, global average pooling to 64 features and a 1×1×1 convolution over one-year age bins; the prediction is the expected value of the bin distribution. We replaced the UK Biobank head (ages 42–82) with 72 bins for 18–90, warm-starting overlapping bins, and fine-tuned with Gaussian soft labels (σ=2 years) and a KL loss: the head for 5 epochs (learning rate 10^{-3}), then the whole network with AdamW [44] (head 10^{-3}, backbone 10^{-4}), early stopping on validation MAE (patience 15). The checkpoint at epoch 30 was frozen. Bias correction followed [3]: $\hat{y}=a\,y+b$ fitted on validation (a=0.832, b=8.30), corrected prediction $(\hat{y}-b)/a$.

Cross-validated replication. To estimate group effects on more than 18 held-out IOP subjects without touching the test set, the 412 train+validation subjects were split into 5 site-stratified folds; each fold was predicted by a model trained on the other four (85% training, 15% inner validation for early stopping and for that fold's bias correction). Full fine-tuning was infeasible on our CPU hardware, so the first three blocks were kept at the UK Biobank weights, which never saw IXI, and the remaining blocks and head were trained with the same objective and schedule. Every non-test subject thus receives an out-of-sample prediction (52 IOP). Cross-validated estimates therefore describe this training recipe applied to IXI, not the frozen main model itself; the main model's held-out estimates are reported alongside them. Training was repeated with 3 seeds, the most our CPU budget allowed (each seed is 5 fold models): the first seed is the primary analysis (as fixed in advance), the other two test whether conclusions depend on initialisation, and the 15 fold models' average forms an ensemble for the locked test set.

B. Analysis plan

Fixed before the test set was evaluated: the split, the exclusions, the bias-correction procedure, the attribution layer and method, and the regression model for group effects. The equivalence margin, the graded verdict rules and the passport criteria below were fixed *after* the test-set analysis had been run, and therefore after the IOP offset had been seen, but before the cross-validation results existed; they were committed to the public repository with a timestamp. They are thus pre-specified only for the cross-validated analysis. The remaining analyses are tests of explanations for an effect already observed and are reported as such.

C. Group fairness and graded verdicts

Group effects were estimated with outcome ~ sex+site+age (OLS, HC3 errors [45]) for corrected BAG and absolute error. Regression effects are reported with HC3 normal-approximation 95% CIs throughout; bootstrap CIs [46] are used only for MAE, r, probe accuracy, matched-pair differences and zero-shot estimates, as marked. The equivalence margin is ±2 years, the order of the scan–rescan variability of brain-age estimates: mean absolute test–retest differences of 1.0–2.3 years were reported for well-performing packages [47], so smaller group differences are not distinguishable from measurement noise in a single scan. Verdicts follow fixed rules: PASS if equivalence is shown by two one-sided tests [48] (90% CI within ±2); INCONCLUSIVE if the 95% CI includes zero without equivalence; CONCERN if the 95% CI excludes zero but the point estimate lies within ±2, or if the effect does not replicate; FAIL if the 95% CI excludes zero, the point estimate lies outside ±2 and the effect replicates (same sign, 95% CI excluding zero) in an independent sample. “Exceeds the margin” refers to the point estimate. The cross-validated estimate (412 subjects) determines the verdict; the main model's test-set estimate serves as replication, and vice versa for the test-set rows. We report the minimum detectable effect at 80% power and Holm-adjusted [49] p-values across all group comparisons. Leave-one-site-out models (trained without one site) measured generalisation to an unseen site.

D. Robustness of the site effect to age

The IOP-vs-Guys effect was re-estimated under: (a) no bias correction; (b) correction fitted on validation; (c) correction fitted on train; (d) age as a covariate of the raw prediction [22]; (e) a B-spline of age (4 df); (f) Guys and HH restricted to IOP's age range; (g) 1:1 age- and sex-matched IOP–Guys pairs (caliper 3 years); and (h) native brain volume

added as a covariate, because registration removes head size from the input. We also inspected gaps of the youngest subjects, near the lower edge of the label range.

E. Is the offset learned?

The UK Biobank model never saw IXI, so its predictions for all subjects aged 42–82 (its output range) show whether the offset exists before fine-tuning. In the cross-validation runs, the gaps of training and out-of-sample IOP subjects were recorded at every epoch.

F. Acquisition sensitivity

Site probe. A class-balanced multinomial logistic regression [26] was trained on train subjects' 64-d features with age, age² and sex regressed out and scored on held-out subjects (balanced accuracy), against a null from 30 label permutations.

Stress test. Each held-out brain's intensity distribution was quantile-matched [31] to each site's reference and re-predicted. References came from train subjects and were standardised to a common age and sex mix (per-site regression of the quantile vector on age, age² and sex, evaluated at the pooled mean), because tissue contrast itself carries age information. Effects are relative to matching to the subject's own site. Contrast, noise, blur and non-uniformity were also varied singly on 40 subjects. The test was repeated with the cross-validated models.

G. Explaining the disparity

Image properties. We compared native resolution and three sharpness measures (gradient, Laplacian, smoothness FWHM [50]) across sites, and blurred each held-out Guys and HH brain to IOP's adjusted smoothness.

Removing site from the features. Site was removed from the 64-d features, with parameters from train only, by ComBat [18], [28] (linear age, age², sex preserved), ComBat-GAM-style harmonisation [29] (spline age), and LEACE concept erasure [27], both plain and conditional (erasing site only from the part of the features not explained by age and sex). Age was re-predicted through the model's own head, which reproduces the model exactly on the original features.

Cohort characteristics. IXI records education, ethnicity, height and weight. We compared sites on university degree, ethnicity and BMI and added them as covariates to the site model, using the zero-shot, main and cross-validated predictions. We also report the ethnicity effect itself.

H. Attribution and its validity

Standard Grad-CAM [32] produced empty maps for many subjects, so we used location-wise gradient × activation (HiResCAM-style [33]) at block `conv_2` (8 mm cells), chosen before any group comparison. Maps were summarised by Harvard–Oxford region [51], both as shares and as density (share divided by the region's share of brain volume), with FDR correction [52]. Validity: a cascading randomisation check [15]; a deletion test [16], [35] on 60 held-out subjects removing top-, bottom- or randomly-ranked 5/10% of cells with two baselines (heavy smoothing, and the brain's mean intensity, which involves no blur), reporting signed and absolute changes; and agreement with Integrated Gradients [34] (zero baseline, 16 steps) on 20 subjects.

I. Uncertainty

We assessed the model's own predictive distribution (coverage and width of its central 50/80/90/95% ranges, per site), split-conformal intervals from out-of-fold absolute residuals [39], pooled and per site (Mondrian [38]), and the spread of the seed ensemble [37]. Training with σ=2 soft labels sets a floor of about 2 years on the model's own spread.

V. RESULTS

A. Accuracy

The UK Biobank SFCN without fine-tuning had a validation MAE of 10.21 years (r=0.910), because its head cannot predict outside 42–82. Fine-tuning reduced it to 3.90. On the test set, evaluated once, the MAE was **3.71** years (Table II). The cross-validated models, which fine-tune only the upper blocks, were less accurate (out-of-sample MAE 4.76 years; 4.76–5.12 across seeds). An independently fine-tuned SFCN and DeepBrainNet reached test MAE 4.07 and 4.53.

TABLE II
Accuracy (95% bootstrap CI)

Model and data	MAE (years)	r
Main model, test (raw)	**3.71** [3.10, 4.38]	0.966
Main model, test (bias-corrected)	3.89 [3.20, 4.61]	0.966
CV models, 412 out-of-sample	4.76	0.943
CV ensemble (15 models), test	5.06	0.949
Zero-shot SFCN, validation	10.21	0.910

B. Group fairness

Table III lists every group effect with its verdict. With 73 test subjects, the sex difference in BAG cannot be shown equivalent (INCONCLUSIVE); with all 412 cross-validated predictions it can (PASS). IOP's BAG offset is −5.24 years in cross-validation (Holm $p<0.001$) and replicates on the test set (−5.06), giving FAIL. HH shows a smaller offset within the margin (CONCERN). The IOP offset was similar for all three seeds (−5.24, −6.23, −5.57) and for the 15-model ensemble on the locked test set (−6.67); the sex difference was not significant for any seed. The test-set CI for IOP is the HC3 interval; an earlier draft reported a bootstrap interval ([−8.66, −1.34]), and we now use HC3 for all regression effects. A leave-one-site-out model never trained on IOP was 3.30 years [1.62, 4.93] less accurate there, whereas one never trained on HH generalised (+0.25).

TABLE III
Group effects (years) and graded verdicts

Effect	Data	Estimate [95% CI]	MDE	Verdict
Sex, BAG	test	−0.59 [−2.79,	3.1	INCONCL.

		1.61]		
Sex, error	test	−0.64 [−1.94, 0.66]	1.9	PASS
HH, BAG	test	+1.06 [−1.48, 3.60]	3.6	INCONCL.
HH, error	test	+0.19 [−1.09, 1.48]	1.8	PASS
IOP, BAG	test	−5.06 [−8.99, −1.13]	5.6	FAIL
IOP, error	test	+1.92 [−1.50, 5.33]	4.9	INCONCL.
Sex, BAG	CV	+0.61 [−0.50, 1.72]	1.6	PASS
Sex, error	CV	+0.11 [−0.58, 0.80]	1.0	PASS
HH, BAG	CV	−1.51 [−2.64, −0.38]	1.6	CONCERN
IOP, BAG	CV	−5.24 [−7.56, −2.92]	3.3	FAIL
IOP, error	CV	+4.06 [2.26, 5.86]	2.6	CONCERN

CV rows decide verdicts; test rows replicate. CV: training recipe, not main model.

C. The IOP offset is not an artefact of age handling

Across all eight age-handling choices, the main model's held-out IOP offset ranged from −3.11 to −4.53 years, with every CI excluding zero (Fig. 3); the weakest estimate came from 1:1 age-matched pairs. In the cross-validated predictions it ranged from −3.94 to −5.24 years, all with $p<0.001$. Native brain volume did not differ between IOP and Guys after age and sex adjustment ($p=0.38$) and did not change the offset. The offset is not at the lower edge of the label range: held-out IOP subjects under 30 had a mean gap of +0.5 years, those over 30 of −5.6 years.

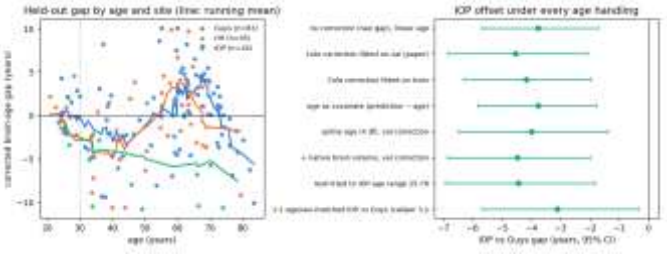


Fig. 3. Main model, held-out subjects. Left: corrected brain-age gap by age and site (lines: running means). Right: the IOP-vs-Guys offset under eight age-handling choices (95% CI).

D. The offset exists before fine-tuning

The UK Biobank model, which never saw an IXI scan, already predicts IOP brains **3.34 years younger** than Guys brains of the same age and sex [−4.71, −1.91] ($p=5\times10^{-6}$, 241 subjects aged 42–82, 19 IOP). Because this model cannot predict below 42, IOP's younger subjects could be pushed towards the floor; but the offset is unchanged with a B-spline for age (−3.55) and in 19 age- and sex-matched pairs (−3.63 [−5.47, −1.74]), and grows when the analysis is restricted to ages 50–80, away from the floor (−5.49 [−6.75, −4.27]; 12 matched pairs: −5.94; Table V). Memorisation of IOP training subjects therefore cannot be its cause, and neither can the output floor. Cross-validated training shows how fine-tuning interacts with it (Fig. 7): after the first epoch, training and out-of-sample IOP subjects both have an age-adjusted offset of about −2.5 to −2.9 years; once the upper blocks start training (stage 2) the two diverge, and by the end of training the offset of the IOP subjects the model trains on has been fitted away (+0.19), while that of unseen IOP subjects has grown to −4.49. Fine-tuning hides the offset for the subjects it sees rather than causing it, which explains why it is absent in the training set but present in every held-out split.

E. The model is sensitive to acquisition appearance

Site is encoded. The probe identified the site of held-out scans with **91.2%** [86.6, 95.2] balanced accuracy after removing age and sex, against a permutation null with mean 34% and 95th percentile 50% (Table IV). This is expected: site is decodable with 99.6% from input intensity quantiles alone and with 92.8% from the pretrained network, so the features inherit an input-level signature; the question is whether the prediction uses it.

TABLE IV
Site probe on held-out subjects (balanced accuracy, %)

Features	Accuracy
Age and sex only	40.8
Permutation null (mean / 95th pct.)	34 / 50
Input intensity quantiles	99.6
Pretrained SFCN (never saw IXI)	92.8
Fine-tuned SFCN, age/sex removed	**91.2**
Cross-validated models (within fold)	86.2

Appearance moves the prediction. Relative to matching to their own site, HH brains given IOP's age- and sex-standardised appearance became **3.92 years younger** [−4.47, −3.38], all 45 of them, and IOP brains given HH's appearance **3.14 years older** (18 of 18; Fig. 4). Between Guys and IOP the effect was small (Table V). With the cross-validated models, the direction replicated in all five fold models over all 412 out-of-sample subjects, with larger magnitudes (HH→IOP: −6.09 years, 99% of subjects). These lighter fine-tunes also responded to the Guys→IOP swap (−3.06), so full fine-tuning appears to reduce, but not remove, acquisition sensitivity.

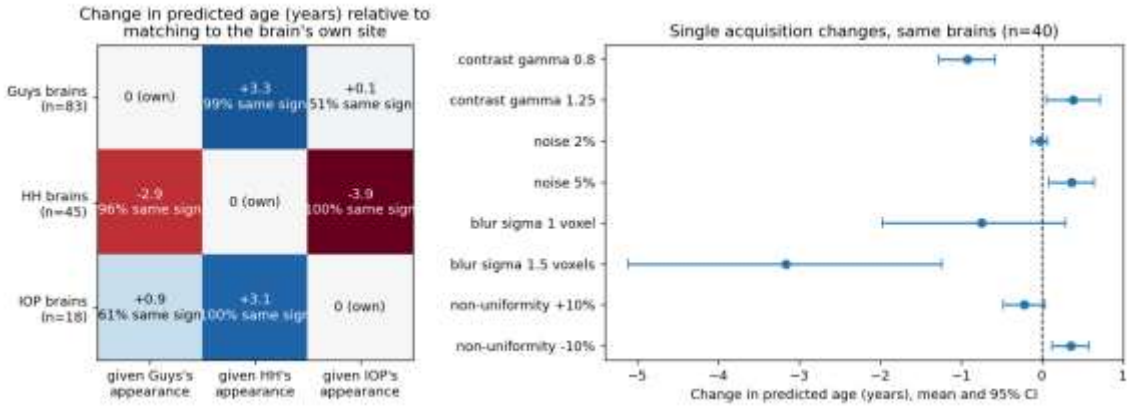


Fig. 4. Acquisition stress test. Left: change in predicted age when each site's held-out brains are given each site's age- and sex-standardised intensity appearance, relative to matching to their own site (share of subjects changing in the majority direction). Right: single acquisition changes on 40 subjects.

F. But acquisition sensitivity does not explain the offset

Table V collects every explanation tested. *Intensity:* swapping between the Guys and IOP appearances moves predictions by at most about 1 year; without age standardisation of the references this effect was overstated (−0.60/+1.60), because IOP's younger recruitment leaked into its reference. *Smoothness:* native resolution is identical across sites, yet IOP inputs are smoother (FWHM +0.87 mm); blurring Guys and HH brains to IOP's smoothness did not lower their predictions. *Removing site from the features:* ComBat erased the site signature (probe 36.3%) with unchanged accuracy, but the offset remained (−3.95), as it did with spline-age ComBat and with age-conditional LEACE. Only *plain* LEACE removed it (+0.40). It did so without lowering overall accuracy or the age–prediction correlation (r=0.962 vs 0.965; Table V), but it moved predictions by 1.1 years on average (conditional LEACE: 0.26) and raised IOP's error from 5.2 to 6.1 years. Because IOP's training subjects are younger, the site direction it erases is partly an age direction for IOP; the conditional version, which first removes age and sex, leaves the offset in place. Spline-age ComBat enlarged the offset (−6.80 with 4 df), but this estimate was unstable across spline flexibility (−6.07, −6.27 and −4.72 with 3, 5 and 6 df) and IOP's error rose with it, so we attribute it to the harmonisation model fitting IOP's narrow, young age range poorly rather than to a larger true offset. *Cohort:* IOP did not differ from Guys in university degree, ethnicity or BMI (adjusted p>0.4), and adjusting for all three left the offset unchanged in the zero-shot, main and cross-validated models. Ethnicity itself showed at most a small effect: non-White participants had a gap +1.43 years higher in cross-validation ([−0.02, 2.88], 74 non-White) and +1.95 in the main model's held-out subjects (p=0.04, 21 non-White), in line with the small ethnicity effects reported elsewhere [20]. Under the stated rule the cross-validated estimate decides, and its CI includes zero without equivalence, so the verdict is INCONCLUSIVE.

TABLE V
Explanations tested for the IOP offset (years)

Test (main model, held-out)	IOP offset	MAE	r
Reference: corrected BAG	−4.53 [−7.04, −2.01]	3.81	0.965
Guys↔IOP appearance swap	+0.11 / +0.90	–	–
Guys / HH blurred to IOP	+0.93 / +0.01	–	–
+ native brain volume	−4.47 [−6.86, −1.98]	–	–
+ degree, BMI, ethnicity	−5.26 [−7.65, −2.86]	–	–
ComBat (linear age)	−3.95 [−6.39, −1.52]	3.90	0.962
ComBat, spline age (4 df)	−6.80 [−10.11, −3.49]	4.01	0.957
ComBat, spline 3 / 5 / 6 df	−6.07 / −6.27 / −4.72	–	–
LEACE, age-conditional	−4.46 [−7.20, −1.73]	3.88	0.962
LEACE, plain	+0.40 [−3.01, 3.82]	3.94	0.962
Zero-shot, linear age, 42–82	−3.34 [−4.71, −1.91]	–	–
Zero-shot, B-spline age, 42–82	−3.55 [−4.94, −2.25]	–	–
Zero-shot, matched pairs, 42–82 (n=19)	−3.63 [−5.47, −1.74]	–	–
Zero-shot, linear age, 50–80	−5.49 [−6.75, −4.27]	–	–
Zero-shot, matched pairs, 50–80 (n=12)	−5.94 [−7.79, −3.64]	–	–

HC3 CIs; zero-shot rows: raw prediction, bootstrap CIs.

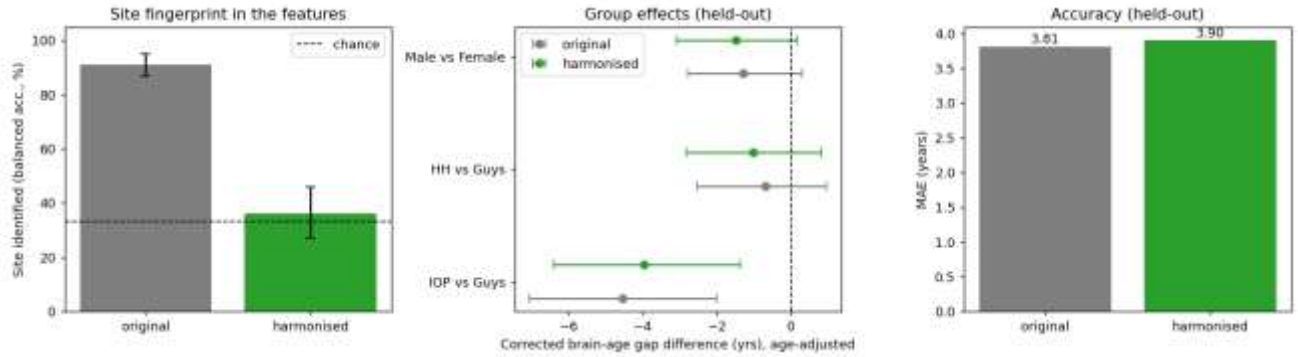


Fig. 5. Main model, held-out subjects. ComBat harmonisation of the features: the site fingerprint is removed (left) and accuracy is unchanged (right), but the IOP offset remains (middle).

G. Where the model looks, and whether to trust it

Attribution concentrates on deep grey matter around the ventricles, periventricular white matter and the cerebellum; normalised by volume, deep grey matter receives 3.2 times its volume share (Fig. 6). The maps depend on the learned weights (randomisation ρ=0.13). Removing the top-ranked 5% of cells changed predictions far more than random or bottom-ranked cells under both baselines (Table VI): $|\Delta\hat{y}|$ 4.83 vs 1.18 vs 0.46 years with smoothing and 6.49 vs 1.99 vs 0.97 with mean infill (top > random in 95% and 88% of subjects, Wilcoxon $p<10^{-8}$; random > bottom, $p<10^{-4}$). The signed change from removing top-ranked cells was positive (+1.67 and +2.56 years): predictions became *older*, opposite to the effect of blur, so the result is not driven by the smoothing baseline. Integrated Gradients agreed with the maps (median Spearman 0.58 across brain cells; 40% overlap of the top 10% of cells, against 10% expected by chance). No region differed by sex after FDR correction. By site, HH and IOP place 13–16% less attribution density on deep grey matter and 12–15% less on the hippocampus and amygdala than Guys ($q<0.001$); these are relative differences in density, not artefacts of region size.

TABLE VI
Deletion test, 60 held-out subjects (years)

Baseline	Removed	Top-ranked	Random	Bottom-ranked
Smoothing	5%	**4.83** (+1.67)	1.18 (+0.22)	0.46 (−0.12)
Smoothing	10%	**6.05** (+0.93)	1.85 (−0.35)	0.71 (−0.05)
Mean infill	5%	**6.49** (+2.56)	1.99 (−0.60)	0.97 (−0.87)
Mean infill	10%	**8.15** (+2.19)	3.32 (−1.23)	1.49 (−1.24)

Mean $|\Delta\hat{y}|$ (mean signed $\Delta\hat{y}$).

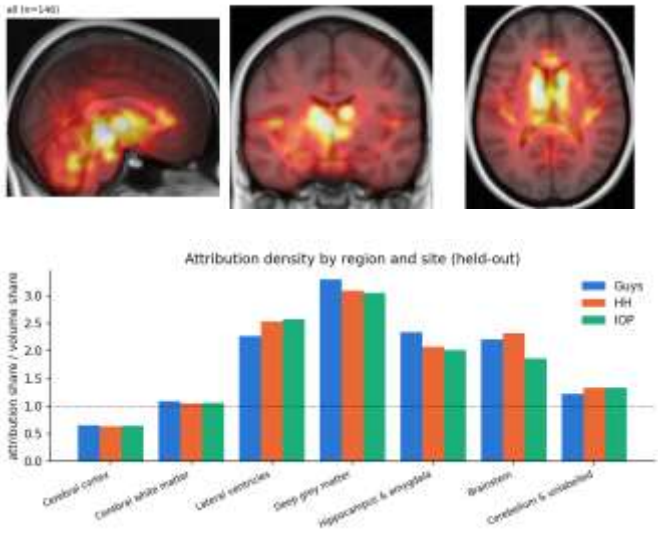


Fig. 6. Top: mean attribution over held-out subjects. Bottom: attribution density (share of attribution divided by share of brain volume) by region and site.

H. Uncertainty

Out of fold (Table VII), the cross-validated models' own 90% range covered 94% of Guys, 89% of HH but only **77%** of IOP subjects. Conformal intervals were calibrated and evaluated on disjoint subjects (calibrate on folds 1–3, evaluate on folds 4–5, then swap; all three seeds). Pooled intervals, which guarantee coverage only on average, covered only 69% of IOP subjects, whereas per-site (Mondrian) intervals covered 91–94% at every site, at the price of much wider intervals for IOP (±20.6 vs ±8.4 years for Guys, from only 26 IOP calibration subjects per split). Average coverage thus hides that one site receives unreliable intervals; uncertainty must be calibrated per site, like accuracy and bias.

A separate analysis gives a different picture for the leave-one-site-out model: trained without IOP, its own spread doubled on IOP (12.46 vs 6.13 years) and its 90% range covered 92% of IOP subjects. The difference is expected: a model that has never seen a site widens its distribution there,

whereas the cross-validated models have seen IOP in training and become over-confident on it.

TABLE VII
Coverage of nominal 90% intervals, out-of-sample (CV)

Interval	Guys (n=233)	HH (127)	IOP (52)
Model's own 90% range	0.94	0.89	**0.77**
Split conformal, pooled	0.97	0.89	**0.69**
Split conformal, per site	0.91	0.92	0.94
Calibration subjects per split	117	64	26
Per-site half-width (years)	8.4	11.4	20.6

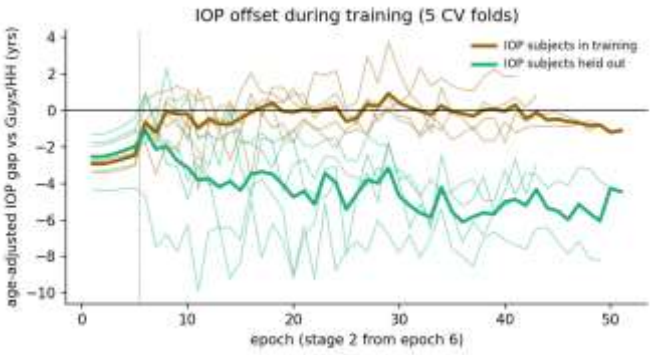


Fig. 7. Age-adjusted IOP offset during cross-validated training (thin: folds; thick: mean). Fine-tuning fits away the offset of IOP subjects in training (brown) but not of unseen IOP subjects (green).

I. Model fairness passport

Table VIII applies the criteria fixed before cross-validation. The full passport, its machine-readable criteria and the script that generates it are released.

TABLE VIII
Model fairness passport (criteria fixed before CV)

Check	Verdict	Evidence
Accuracy	PASS	MAE 3.71, r 0.966
Fairness by sex	PASS	gap and error equivalent within ±2 yrs (CV)
Fairness by site	FAIL	IOP −5.2 yrs (CV), replicated, robust to age; HH CONCERN
Invariance to intensity-swap stress test	FAIL	HH↔IOP appearance shifts 3–4 yrs
Faithful explanations	PASS	ρ 0.13; deletion top > random under both baselines; IG agrees
Honest uncertainty	FAIL	own 90% range covers 77% at IOP; per-site conformal restores 92%

VI. DISCUSSION

A replicated disparity that is not an age artefact. IOP's lower brain-age gap survives every age-handling choice we could construct, appears in independent splits and in cross-validation, and is present before fine-tuning. The usual first explanation of site differences in brain age, different age distributions [20], does not account for it.

Fine-tuning hides the offset where it is easiest to check. Perhaps the most practical finding is Fig. 7: the offset is present in every subject before fine-tuning, and fine-tuning removes it only for the IOP subjects the model is trained on (+0.19 years at the end of training) while it grows for unseen IOP subjects (−4.49). An audit that inspects training-set residuals, or reports in-sample site effects, would conclude that the model is fair across sites. Site effects must be estimated on subjects the model has not seen, and ideally with cross-validation, because a held-out split from a small site is too small to detect them.

Sensitive to acquisition, but the sensitivity is not the explanation. The model is not acquisition-invariant: an unchanged HH brain given IOP's appearance loses almost four years. Yet the explanations we can measure—intensity profile, smoothness, head size, recorded cohort characteristics, and the site signature in the features—each leave the IOP offset in place. Two methodological lessons follow. First, references for intensity and erasure must be conditioned on age, because tissue contrast and the site–age confound carry age information: unconditioned versions overstated the intensity effect and made the offset seem to vanish. Second, removing a site signature is not the same as removing a site disparity. A study that had stopped at the stress test, or at plain erasure, would have reported "scanner bias"; the evidence here does not support that label.

What remains. The offset could stem from acquisition properties not modelled here (e.g., spatially varying contrast or vendor-specific reconstruction) or from cohort characteristics IXI does not record. Within IXI, site and cohort are confounded, so these cannot be separated; travelling-subject data would.

Limitations. IOP contributes 61 subjects in total. The cross-validated models fine-tune only the upper blocks, a lighter recipe than the main model, so their estimates describe that recipe; the main model's held-out estimates agree in sign and size, but a fully fine-tuned cross-validation remains to be run. The equivalence margin and verdict rules were fixed after the test-set analysis, so they are pre-specified only for the cross-validated analysis. Three seeds give a limited view of training variability. Quantile matching changes only the global intensity distribution, and Gaussian blur did not reproduce IOP's full sharpness profile. Attribution has 8 mm resolution, and deletion tests are themselves affected by distribution shift. We did not evaluate a classical feature-based baseline or a second dataset. Participants are healthy volunteers; no clinical conclusion follows.

Future work. Travelling-subject datasets, a FreeSurfer-feature baseline (if it shows the same offset, the cause lies in

the images or cohort rather than a CNN shortcut), site-adversarial fine-tuning [19], and passports for further models and datasets.

VII. CONCLUSION

A fine-tuned SFCN is accurate on IXI and its errors are equivalent between sexes, but it shows a replicated 4–5 year brain-age offset at one site. The offset is robust to how age is handled and exists before fine-tuning. The model is clearly sensitive to acquisition appearance, yet neither that sensitivity nor any measured acquisition or cohort property explains the offset. Site disparities in brain age should be tested, not labelled: a protocol of age-robust estimates, stress tests, age-conditional erasure and graded verdicts shows what can and cannot be concluded.

CODE AND DATA AVAILABILITY

Code, the split and exclusion lists, cross-validation folds, environment, passport criteria and generator, an interactive website and a live demo are at `https://github.com/Jaaiwanth/brain-age-fairness-xai`. IXI is available under CC BY-SA 3.0 [40].

AUTHOR CONTRIBUTIONS

Adharshan N and Jaaiwanth K contributed equally: both designed the study, verified the labels, built the preprocessing and fine-tuning pipelines (including the independent SFCN and DeepBrainNet replication), performed the analyses, built the website, demo and passport, and wrote the manuscript. Shridevi S supervised the project and reviewed the manuscript.

ACKNOWLEDGMENT

We thank the IXI project for the data and the SFCN authors for releasing their pretrained weights.

REFERENCES

[1] J. H. Cole and K. Franke, “Predicting age using neuroimaging: innovative brain ageing biomarkers,” *Trends in Neurosciences*, vol. 40, no. 12, pp. 681–690, 2017.

[2] K. Franke and C. Gaser, “Ten years of BrainAGE as a neuroimaging biomarker of brain aging: what insights have we gained?,” *Frontiers in Neurology*, vol. 10, Art. no. 789, 2019.

[3] J. H. Cole, S. J. Ritchie, M. E. Bastin et al., “Brain age predicts mortality,” *Molecular Psychiatry*, vol. 23, pp. 1385–1392, 2018.

[4] J. H. Cole, R. P. K. Poudel, D. Tsagkrasoulis et al., “Predicting brain age with deep learning from raw imaging data results in a reliable and heritable biomarker,” *NeuroImage*, vol. 163, pp. 115–124, 2017.

[5] H. Peng, W. Gong, C. F. Beckmann, A. Vedaldi, and S. M. Smith, “Accurate brain age prediction with lightweight deep neural networks,” *Medical Image Analysis*, vol. 68, Art. no. 101871, 2021.

[6] V. M. Bashyam, G. Erus, J. Doshi et al., “MRI signatures of brain age and disease over the lifespan based on a deep brain network and 14 468 individuals worldwide,” *Brain*, vol. 143, no. 7, pp. 2312–2324, 2020.

[7] E. H. Leonardsen, H. Peng, T. Kaufmann et al., “Deep neural networks learn general and clinically relevant representations of the ageing brain,” *NeuroImage*, vol. 256, Art. no. 119210, 2022.

[8] A. J. Larrazabal, N. Nieto, V. Peterson, D. H. Milone, and E. Ferrante, “Gender imbalance in medical imaging datasets produces biased classifiers for computer-aided diagnosis,” *Proceedings of the National Academy of Sciences*, vol. 117, no. 23, pp. 12592–12594, 2020.

[9] L. Seyyed-Kalantari, H. Zhang, M. B. A. McDermott, I. Y. Chen, and M. Ghassemi, “Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations,” *Nature Medicine*, vol. 27, pp. 2176–2182, 2021.

[10] M. A. Ricci Lara, R. Echeveste, and E. Ferrante, “Addressing fairness in artificial intelligence for medical imaging,” *Nature Communications*, vol. 13, Art. no. 4581, 2022.

[11] C. Piçarra and B. Glocker, “Analysing race and sex bias in brain age prediction,” arXiv preprint arXiv:2309.10835, 2023.

[12] R. Geirhos, J. Jacobsen, C. Michaelis et al., “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, pp. 665–673, 2020.

[13] J. R. Zech, M. A. Badgeley, M. Liu, A. B. Costa, J. J. Titano, and E. K. Oermann, “Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: a cross-sectional study,” *PLoS Medicine*, vol. 15, no. 11, Art. no. e1002683, 2018.

[14] A. J. DeGrave, J. D. Janizek, and S. Lee, “AI for radiographic COVID-19 detection selects shortcuts over signal,” *Nature Machine Intelligence*, vol. 3, pp. 610–619, 2021.

[15] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, “Sanity checks for saliency maps,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018.

[16] S. Hooker, D. Erhan, P. Kindermans, and B. Kim, “A benchmark for interpretability methods in deep neural networks,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019.

[17] B. Glocker, R. Robinson, D. C. Castro, Q. Dou, and E. Konukoglu, “Machine learning with multi-site imaging data: an empirical study on the impact of scanner effects,” arXiv preprint arXiv:1910.04597, 2019.

[18] J. Fortin, N. Cullen, Y. I. Sheline et al., “Harmonization of cortical thickness measurements across scanners and sites,” *NeuroImage*, vol. 167, pp. 104–120, 2018.

[19] N. K. Dinsdale, M. Jenkinson, and A. I. L. Namburete, “Deep learning-based unlearning of dataset bias for MRI harmonisation and confound removal,” *NeuroImage*, vol. 228, Art. no. 117689, 2021.

[20] L. J. Aguzin Parrilli and M. A. Belzunce, “Bias and generalizability of brain age prediction models: A multi-cohort evaluation with anatomical and interpretability insights,” *Imaging Neuroscience*, vol. 4, Art. no. IMAG.a.1164, 2026.

[21] M. Mitchell, S. Wu, A. Zaldivar et al., “Model cards for model reporting,” in *Proc. Conference on Fairness, Accountability, and Transparency (FAT*)*, pp. 220–229, 2019.

[22] T. T. Le, R. T. Kuplicki, B. A. McKinney et al., “A nonlinear simulation framework supports adjusting for age when analyzing BrainAGE,” *Frontiers in Aging Neuroscience*, vol. 10, Art. no. 317, 2018.

[23] S. M. Smith, D. Vidaurre, F. Alfaro-Almagro, T. E. Nichols, and K. L. Miller, “Estimation of brain age delta from brain imaging,” *NeuroImage*, vol. 200, pp. 528–539, 2019.

[24] A. G. de Lange and J. H. Cole, “Commentary: Correction procedures in brain-age prediction,” *NeuroImage: Clinical*, vol. 26, Art. no. 102229, 2020.

[25] B. Glocker, C. Jones, M. Bernhardt, and S. Winzeck, “Algorithmic encoding of protected characteristics in chest X-ray disease detection models,” *eBioMedicine*, vol. 89, Art. no. 104467, 2023.

[26] G. Alain and Y. Bengio, “Understanding intermediate layers using linear classifier probes,” arXiv preprint arXiv:1610.01644, 2016.

[27] N. Belrose, D. Schneider-Joseph, S. Ravfogel, R. Cotterell, E. Raff, and S. Biderman, “LEACE: Perfect linear concept erasure in closed form,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023.

[28] W. E. Johnson, C. Li, and A. Rabinovic, “Adjusting batch effects in microarray expression data using empirical Bayes methods,” *Biostatistics*, vol. 8, no. 1, pp. 118–127, 2007.

[29] R. Pomponio, G. Erus, M. Habes et al., “Harmonization of large MRI datasets for the analysis of brain imaging patterns throughout the lifespan,” *NeuroImage*, vol. 208, Art. no. 116450, 2020.

[30] Y. Ganin, E. Ustinova, H. Ajakan et al., “Domain-adversarial training of neural networks,” *Journal of Machine Learning Research*, vol. 17, no. 59, pp. 1–35, 2016.

[31] L. G. Nyúl, J. K. Udupa, and X. Zhang, "New variants of a method of MRI scale standardization," *IEEE Transactions on Medical Imaging*, vol. 19, no. 2, pp. 143–150, 2000.

[32] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, pp. 618–626, 2017.

[33] R. L. Draelos and L. Carin, "Use HiResCAM instead of Grad-CAM for faithful explanations of convolutional neural networks," arXiv preprint arXiv:2011.08891, 2020.

[34] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," in *Proc. International Conference on Machine Learning (ICML)*, pp. 3319–3328, 2017.

[35] W. Samek, A. Binder, G. Montavon, S. Lapuschkin, and K. Müller, "Evaluating the visualization of what a deep neural network has learned," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 28, no. 11, pp. 2660–2673, 2017.

[36] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. International Conference on Machine Learning (ICML)*, pp. 1321–1330, 2017.

[37] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.

[38] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. New York: Springer, 2005.

[39] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.

[40] IXI Dataset. [Online]. Available: https://brain-development.org/ixi-dataset/

[41] N. J. Tustison, B. B. Avants, P. A. Cook et al., "N4ITK: Improved N3 bias correction," *IEEE Transactions on Medical Imaging*, vol. 29, no. 6, pp. 1310–1320, 2010.

[42] N. J. Tustison, P. A. Cook, A. J. Holbrook et al., "The ANTsX ecosystem for quantitative biological and medical imaging," *Scientific Reports*, vol. 11, Art. no. 9068, 2021.

[43] R. Ciric, W. H. Thompson, R. Lorenz et al., "TemplateFlow: FAIR-sharing of multi-scale, multi-species brain models," *Nature Methods*, vol. 19, pp. 1568–1571, 2022.

[44] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. International Conference on Learning Representations (ICLR)*, 2019.

[45] J. G. MacKinnon and H. White, "Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties," *Journal of Econometrics*, vol. 29, no. 3, pp. 305–325, 1985.

[46] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. New York: Chapman & Hall, 1993.

[47] R. P. Dörfel, J. M. Arenas-Gomez, P. M. Fisher, M. Ganz, G. M. Knudsen, J. E. Svensson, and P. Plavén-Sigray, "Prediction of brain age using structural magnetic resonance imaging: A comparison of accuracy and test–retest reliability of publicly available software packages," *Human Brain Mapping*, vol. 44, no. 17, pp. 6139–6148, 2023.

[48] D. J. Schuirmann, "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability," *Journal of Pharmacokinetics and Biopharmaceutics*, vol. 15, no. 6, pp. 657–680, 1987.

[49] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[50] S. D. Forman, J. D. Cohen, M. Fitzgerald, W. F. Eddy, M. A. Mintun, and D. C. Noll, "Improved assessment of significant activation in functional magnetic resonance imaging (fMRI): use of a cluster-size threshold," *Magnetic Resonance in Medicine*, vol. 33, no. 5, pp. 636–647, 1995.

[51] M. Jenkinson, C. F. Beckmann, T. E. J. Behrens, M. W. Woolrich, and S. M. Smith, "FSL," *NeuroImage*, vol. 62, no. 2, pp. 782–790, 2012.

[52] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B*, vol. 57, no. 1, pp. 289–300, 1995.